

[DC] Exploring Environmental Customization in Virtual Reality Mindful Movement to Reduce Anxiety and Stress in University Students

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ABSTRACT

University students face high stress and anxiety, impacting mental health and academic performance. This study investigates the effects of customizable virtual reality environments during mindful movement exercises on reducing stress and anxiety. Grounded in Self-Determination and Stress-Coping Theories, the research includes three phases: evaluating psychological and physiological outcomes, exploring user experiences, and collaborating with stakeholders to integrate VR into student wellness programs. Findings aim to demonstrate the benefits of VR customization for mental well-being, offering scalable solutions for student wellness and advancing VR applications in mental health.

Index Terms: Virtual reality, health informatics, user interface customization

1 RESEARCH QUESTION

How does customizing virtual environments during structured mindful movement affect anxiety and stress reduction in university students compared to pre-set non-customizable environments?

2 BACKGROUND AND RATIONALE

University students often experience high levels of stress and anxiety [1,12], which can negatively impact mental health [14,9] and academic performance [9]. Engaging in mindful movement exercises (e.g., yoga, stretching) is recognized as an effective way to reduce stress and anxiety [7,2,8], but many students find it difficult to incorporate a mindful movement practice into their routines, often due to a lack of motivation or access to appropriate or comfortable environments [4,5,3]. Virtual Reality (VR) has emerged as a promising tool to create immersive environments for exercise, potentially simulating settings that promote psychological well-being [10,11]. Customizing these VR environments, by adjusting visual elements such as season, location, weather, or time of day, could provide further psychological benefits by allowing users to control the setting to align with their comfort and preferences. However, little is known about how VR customization can specifically target anxiety and stress reduction among university students. This research aims to explore the effects of customizable VR environments on reducing anxiety and stress levels in university students and how these findings can inform university health and wellness programs.

2 THEORETICAL APPROACH

The research will be grounded in Self-Determination Theory, which suggests that autonomy is a key factor in fostering intrinsic motivation and psychological well-being [13]. Enabling students to

customize their VR environments could increase their sense of autonomy, potentially leading to better mental health outcomes. Additionally, Stress and Coping Theory suggests that environments perceived as safe, relaxing, and under one's control can reduce stress and anxiety [6].

4 OBJECTIVES

This research will address the following objectives through three interconnected research phases: 1) Compare the psychological and physiological effects of customizable vs. pre-set VR environments on anxiety and stress during mindful movement exercise; 2) Explore university students' subjective experiences with customizable VR environments and their impact on mental well-being; 3) Collaborate with key stakeholders to develop strategies for integrating customizable VR exercise environments into university wellness programs to reduce student anxiety and stress.

4.1 Phase One

This study will examine the psychological and physiological effects of customizable VR environments on anxiety and stress during mindful movement exercise in 100 university students. Diversity in terms of gender, year of study, GPA, and baseline anxiety levels will be considered during recruitment to ensure a representative sample. Participants will be randomly assigned to either a customizable or pre-set VR environment for a 20-minute stretching session. Psychological measures (self-reported anxiety, stress, and mood) will be taken before and after the session, while sensor-derived physiological data (heart rate variability, respiratory rate, and electrodermal activity) will measure stress responses during the session. A mixed design ANOVA will assess the effects of time (pre- and post-session) and group (customizable vs. pre-set VR) on psychological measures. Repeated measures ANOVA will assess changes in physiological measures over time during the session.

4.2 Phase Two

This study will qualitatively explore university students' experiences with customizable VR environments during mindful movement exercise, focusing on their impact on anxiety, stress, and motivation for continued use. A sample of 15 students from Phase 1 will participate in semi-structured interviews. Topics will include how customization affects anxiety, stress, the perceived effectiveness of creating a relaxing atmosphere, and likelihood of continued VR-based mindful movement. Interviews will be transcribed and analyzed using thematic analysis to identify patterns related to the influence of customization on anxiety and stress reduction.

4.3 Phase Three

This study focuses on knowledge translation through a Community Action Partnership (CAP) to develop strategies for integrating customizable VR exercise environments into student wellness programs. The CAP will include 5-10 key stakeholders from the community, such as participants from Phase 2, university health

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committee members, fitness staff, student services, and mental health professionals. Together, the CAP will first create an infographic based on findings from Phases 1 and 2, outlining the benefits and guidelines for customizable VR use. Next, a focus group with the CAP will be facilitated to review the resource for clarity and practicality, with feedback used to refine it. Ongoing input from CAP members will ensure effective implementation and communication strategies.

5 EXPECTED IMPACT

This research will enhance both theoretical knowledge and practical application of customizable VR exercise environments in reducing stress and anxiety. By collaborating key stakeholders through the CAP, the study aims to develop actionable strategies for integrating VR into university wellness programs, ensuring these interventions are both relevant and sustainable. The findings are anticipated to show that customizable VR environments can effectively reduce anxiety and stress among university students during mindful movement, with broader potential applications in public health. These could include rehabilitation (such as physical, cardiac, neurocognitive, social cognitive, and offender rehabilitation), and treatments for conditions like post-traumatic stress disorder and phobias. The results will be disseminated via academic publications and relevant conferences. An infographic, co-developed with the CAP, will outline the benefits of VR environments and provide guidelines for their use in wellness programs. The resource will be shared with university fitness and wellness centers to encourage VR-based initiatives, with ongoing stakeholder collaboration to ensure practical implementation and refinement.

6 PROGRESS REPORT

To date, I have developed a detailed outline of my proposed research, conducted a thorough literature review, and completed required competencies in statistics. The remaining research tasks involve the development of the VR environments and the execution of the three study phases. This includes participant recruitment, data collection, and comprehensive analysis. The primary focus will be on implementing the research plan and synthesizing the findings to address the study objectives. Remaining degree requirements include dissertation proposal (April 2025), and dissertation submission and defense (December 2026).

7 CONCLUSION

This research investigates the potential of customizable VR environments to enhance the effectiveness of mindful movement exercises in reducing anxiety and stress among university students. By integrating user-driven customization, this research aims to bridge gaps in current VR-based wellness strategies and contribute to both theoretical and practical advancements in the field. Looking ahead, this work raises several questions that merit further exploration. Key questions include: (1) how can real-time biometric feedback be incorporated into the current study to dynamically adjust the environment and optimize the user experience?; (2) what strategies can ensure the scalability of VR-based wellness interventions within university programs, considering diversity in user needs and institutional resources?; and (3) how can the sustained effects of customizable VR interventions on stress and anxiety reduction be effectively assessed beyond the initial implementation phase? These areas represent opportunities for mentorship, collaboration, and discourse.

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